

Lecture 14

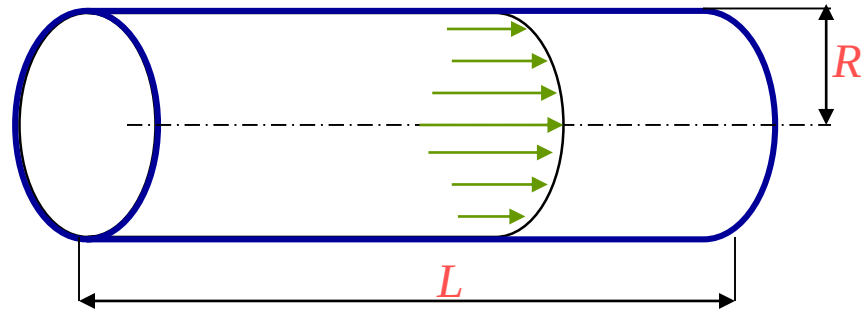
ChE 467/596-012 Polymer Rheology

Saad A Khan

Pressure Driven Flow (Capillary Rheometer)

Principle => pipe flow

Non-homogenous flow



Measure shear stress at wall:

$$\tau_{\text{wall}} = \tau_{12} \Big|_{r=R} = \frac{R}{2} \frac{\Delta p}{L}$$

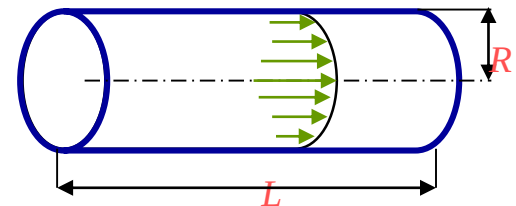
Measure flowrate:

$$\dot{\gamma}_{\text{wall}} = \frac{Q}{\pi R^3} \left[3 + \frac{d \ln Q}{d \ln \tau_{\text{wall}}} \right]$$

$$\eta = \frac{\tau_{\text{wall}}}{\dot{\gamma}_{\text{wall}}}$$

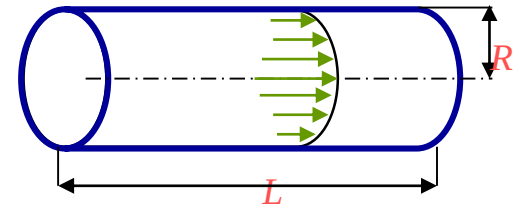
Capillary Rheometers-Disadvantages

- **Non-homogenous flow** (*Correction factor needed*)
- **High shear rates only**
- **Difficult to Clean** (*sample sticks on wall, solidification*)
- **Can Only measure viscosity**
 - *No normal stresses*
 - *No dynamic properties (G' , G'' , etc)*
- **Compressibility issues**
- **Viscous heating**
- **Pressure effect on η**



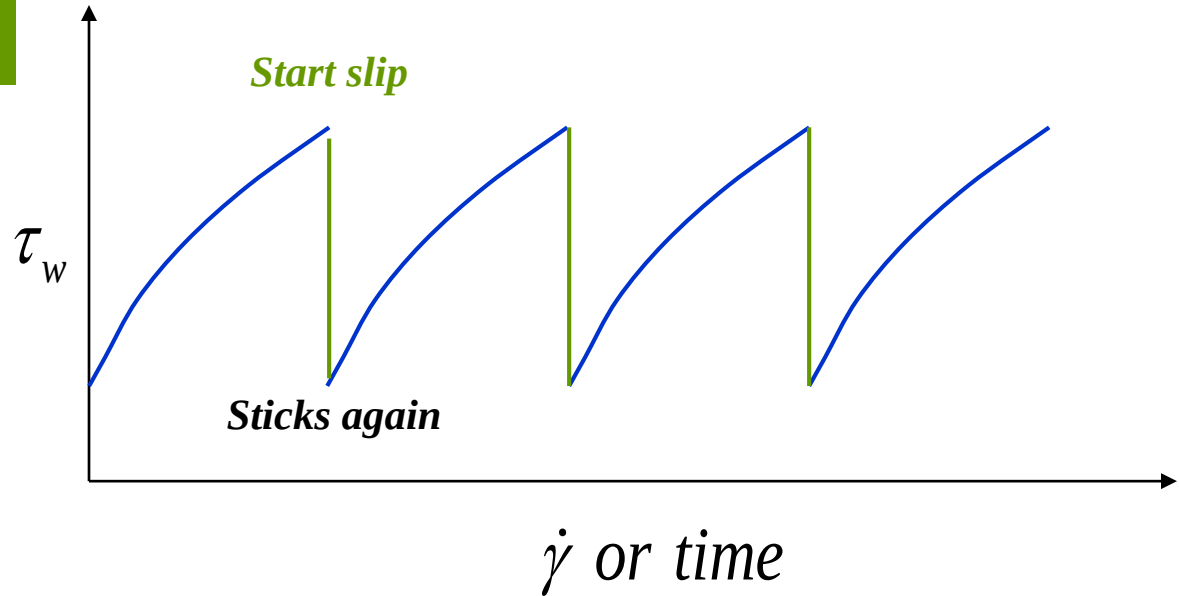
Capillary Rheometer-Advantages

- **No sample evaporation** (*Temperature studies*)
- **Can get high shear rate data**
 - *Relevant in processing*
 - *Expands the shear rate regime*
- **Similar to process flow** (*can have it as online device*)
- **Enclosed** (*good for volatile materials*)
- **Can use high pressure**



Cappillary rheometry –other issues

Stick Slip



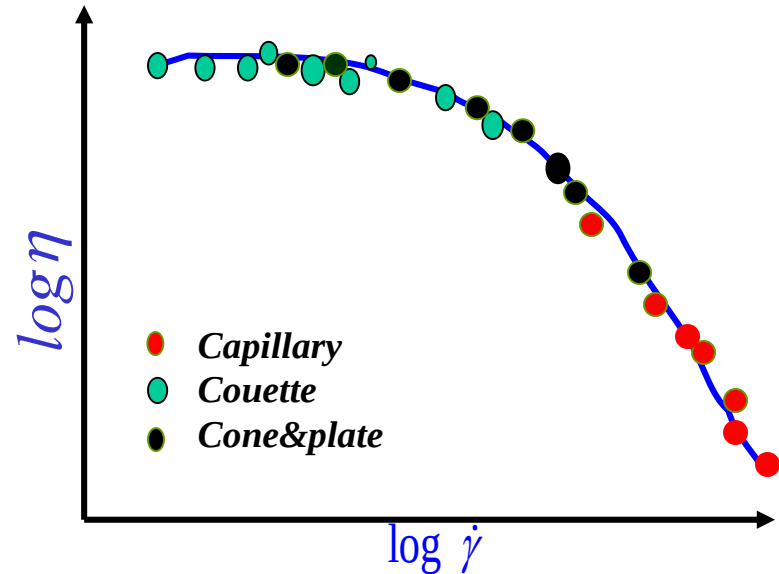
Melt

High shear rates $\rightarrow$ unstable flow $\rightarrow$ extrudate appears distorted

Jagged polymer samples coming out

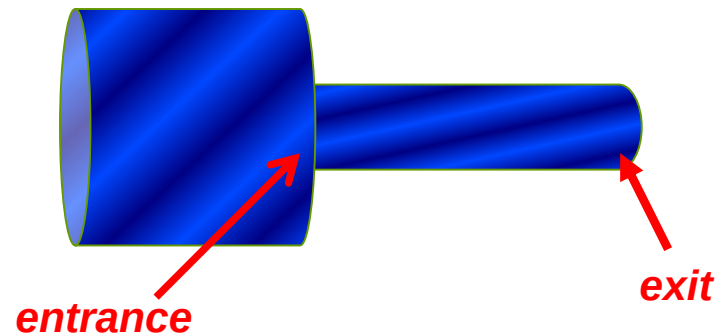
Capillary Rheometry-Other Issues

•Data can be combined with data from couette and cone & plate rheometers to get wide shear rate range



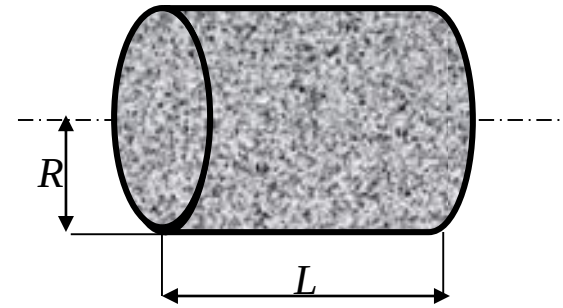
Wall slip – need to correct shear rate

Entrance/exit effects
-need to correct stress



Wall Slip Effects

- *Wall slip can happen, particularly if you have filled systems*
- *Fillers migrate inwards → particle lean layer near wall with much lower viscosity → slip layer*



To check for wall slip:

Measure viscosity at similar L/R ratio but different R

If results overlap => no wall slip

If different results => wall slip present

Wall Slip Effects, cont'd

Correcting shear rate for wall slip

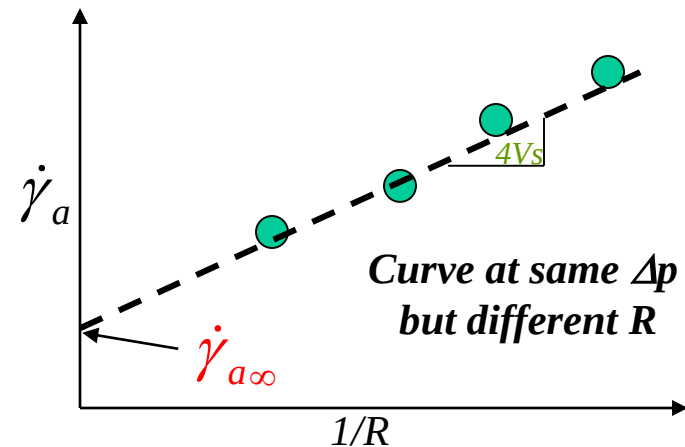
- Pick capillaries of different radii R
- Run experiment with each tube keeping Δp (i.e. τ_{wall}) same
- Plot app. shear rate vs $1/R$
- $\dot{\gamma}_{a\infty}$ is the corrected shear rate

$$\dot{\gamma}_a = \dot{\gamma}_{a\infty} + \frac{4V_s}{R}$$

Get viscosity from $\eta = \frac{\tau_{\text{wall}}}{\dot{\gamma}_{a\infty}}$

$$\dot{\gamma}_a = \frac{Q}{\pi R^3} \left[3 + \frac{d \ln Q}{d \ln \tau_{\text{wall}}} \right]$$

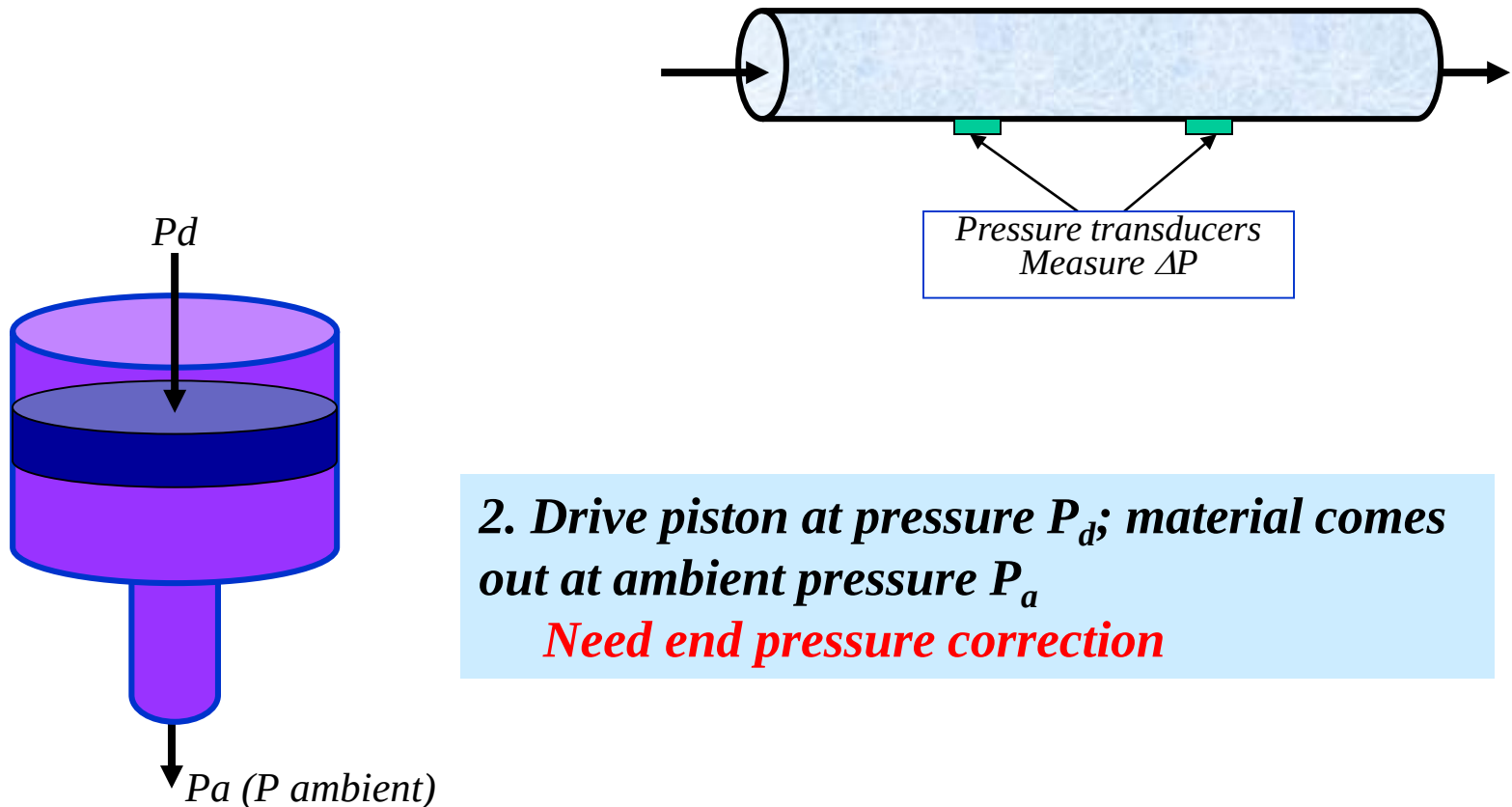
$\dot{\gamma}_a$ = apparent shear rate



- Repeat with different Δp to get different corrected shear rate

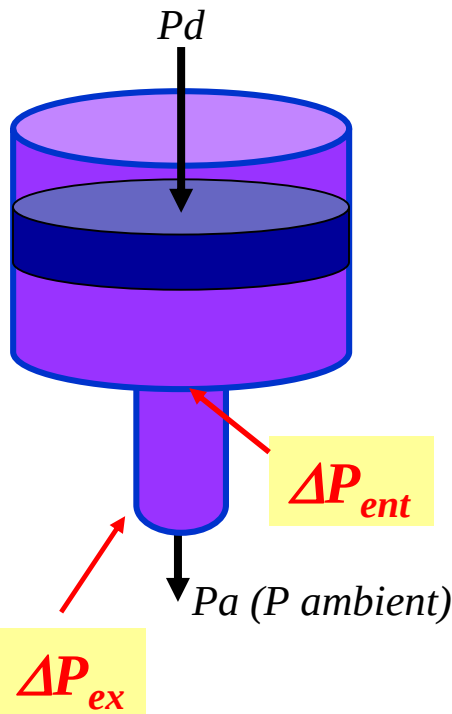
End Pressure Effects

1. Place pressure transducers far from the entrance and exit
No end pressure effect



2. Drive piston at pressure P_d ; material comes out at ambient pressure P_a
Need end pressure correction

End Pressure Effects



Measure $P_d - P_a$

$$\text{But } P_d - P_a = \Delta P_{ent} + \Delta P + \Delta P_{ex}$$

ΔP_{ent}
xs pressure from
entrance flow

ΔP
pressure from flow

ΔP_{ex}
xs pressure from
exit flow

$\Delta P_{end} = \Delta P_{ent} + \Delta P_{ex}$ introduces errors and over predicts stress if not taken into account. How?

Because measured $P_d - P_a$ often mistakenly taken as $\Delta P = 2\tau_w L/R$

End Pressure Effects-Bagley Correction

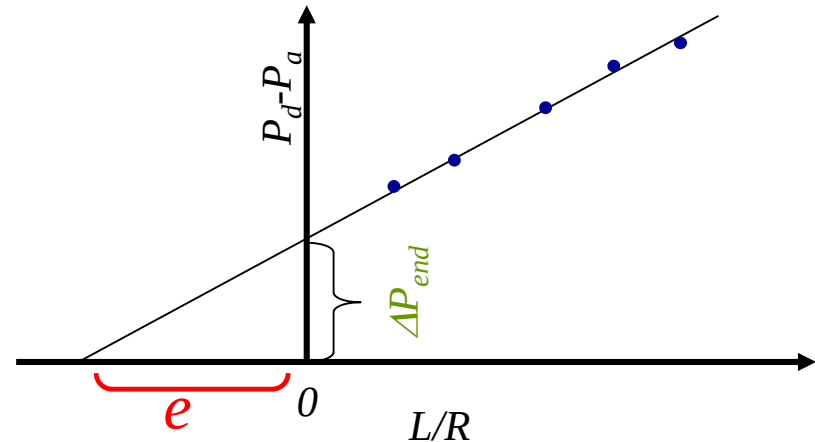
To correct for end pressure effect:

- *Keep Q constant and measure P_d versus R (P_a known)*
- *Plot $P_d - P_a$ vs L/R*
- *From Q get shear rate*
- *Get τ from the equation*

$$P_d - P_a = \frac{2\tau_w L}{R} + \Delta P_{end}$$

OR

$$\tau_w = \frac{P_d - P_a}{2(L/R + e)}$$



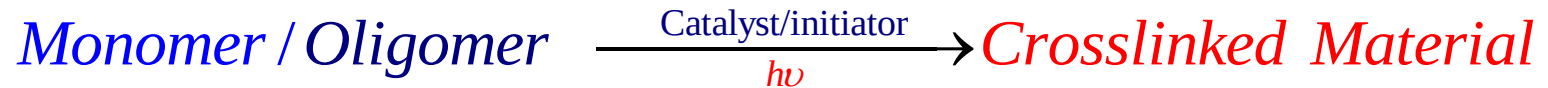
*Ignore subsequent slides on capillary rheometry
Just go to gels*

Gels and Networks

**Probing microstructure, state and kinetics
through rheology**

Gels and networks

Chemically x-linked systems



Physical Gels (reversible)

Temperature, light, hydrogen bonds

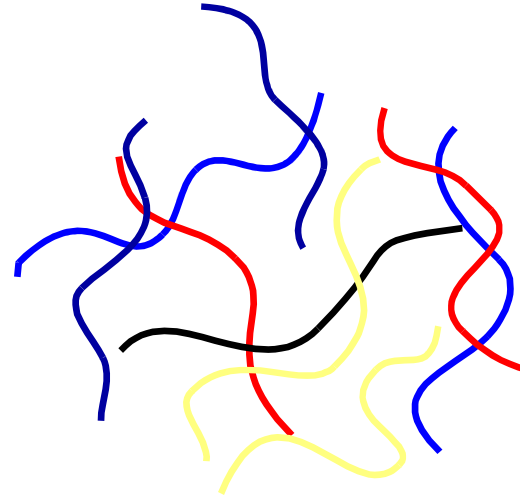
What is a Gel?

3-Dimensional network spanning the sample

Gels and networks, cont'd

Polymer chains:

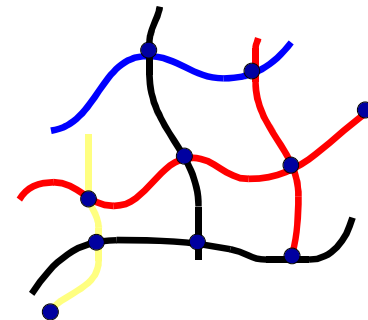
Chains can move/wiggle around through “reptation”



Gels:

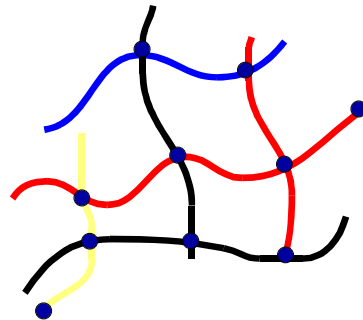
The polymer chains are constrained

Junctions can be fixed chemically or physically



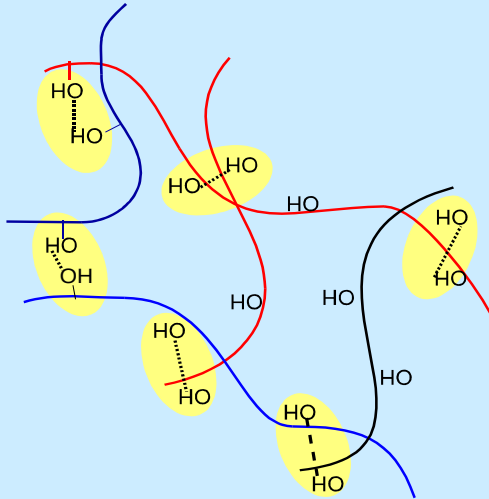
Chemically x-linked gels

Fix Junctions chemically $\Rightarrow$ Covalent bonds
 $\Rightarrow$ Chemically x-linked gels/system
(Chemically cured system/thermosets)

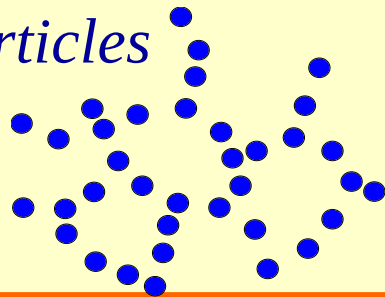


Physical Gel-Examples

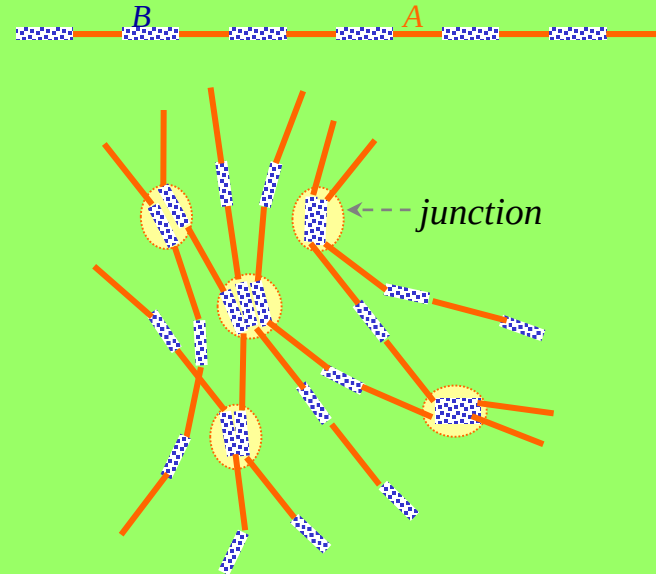
Systems with Hydrogen bonds



Colloidal Particles



Block Copolymers

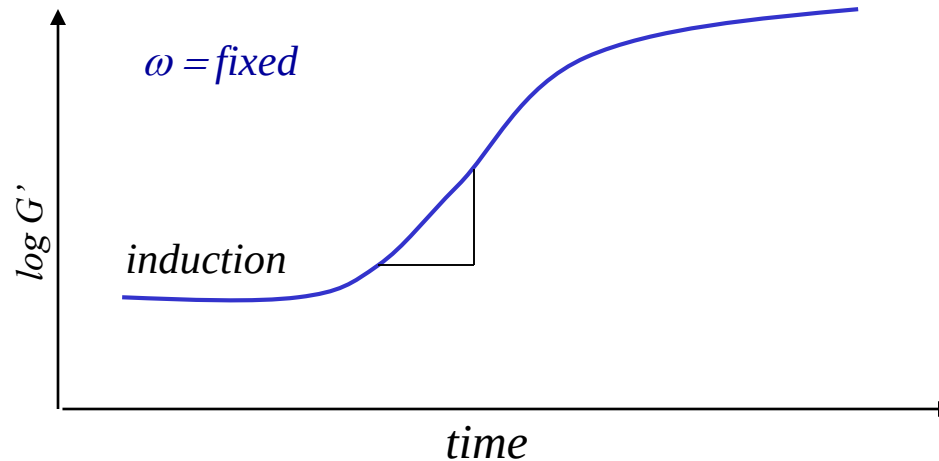


- *Ionic interactions*
- *Hydrophobic interactions*

Critical Issues

1. *What is a good way to monitor gelation/x-linking rheologically?*
2. *How can one differentiate between a gel and a sol/solution?*
3. *How can one get the gel point?*

Monitoring Gelation Kinetics

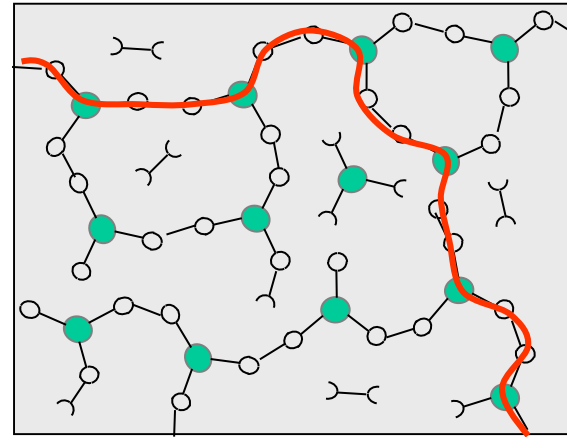
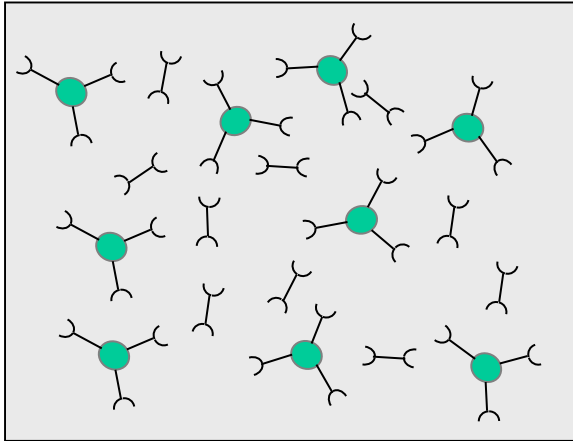


$$G'(t) = n(t) kT + G_{en}$$

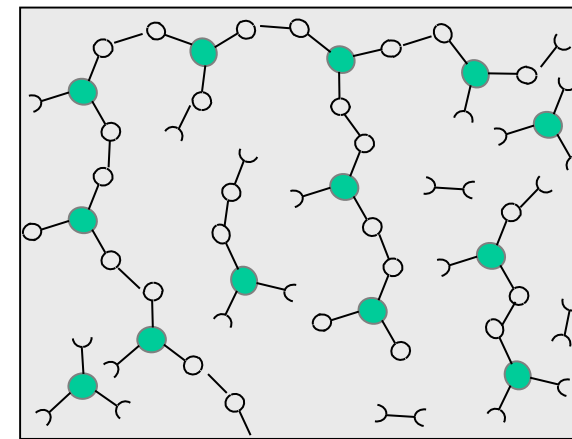
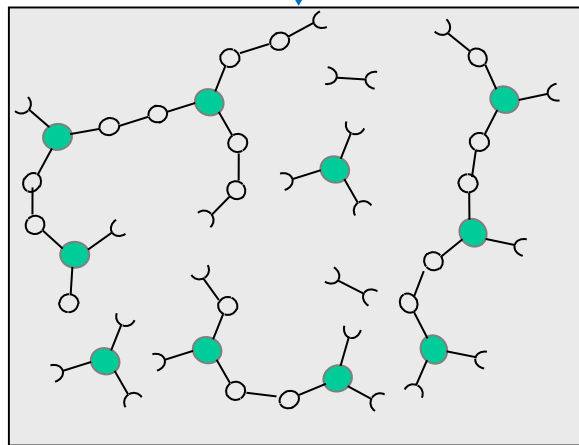
Equivalent to # x-links

$\frac{dG'}{dt}$ is used to relate to kinetics

Microstructure Evolution During X-linking

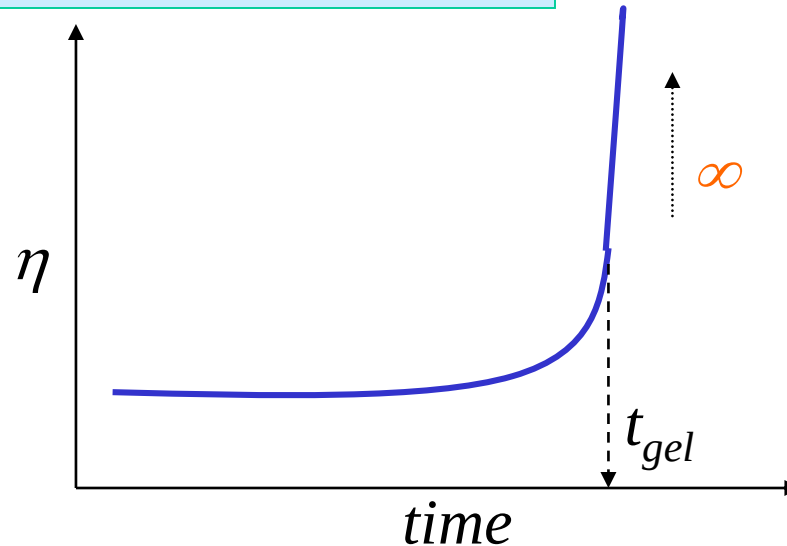


Gel point



Gel point detection – Approach 1

1. Viscosity versus time behavior



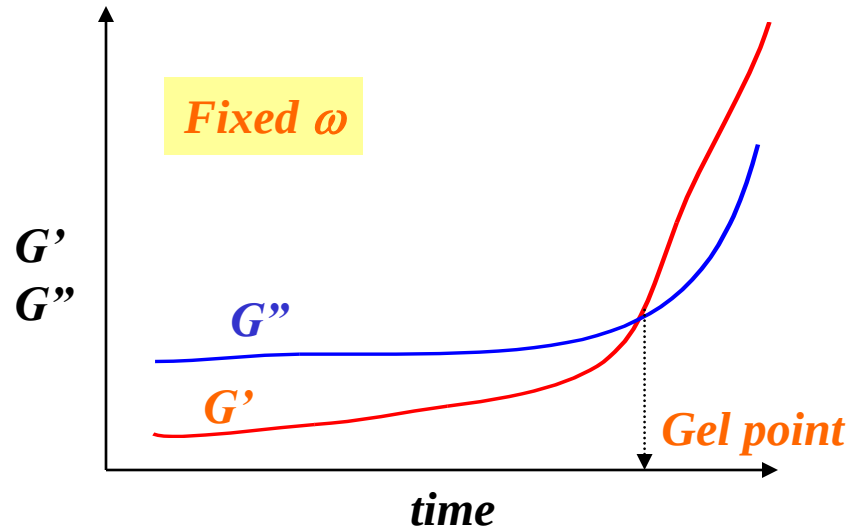
Lecture ended
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Time at which $\eta \rightarrow \infty \Rightarrow$ gelation time $\rightarrow$ Gelation process

Any Problem? shear can interfere with crosslinking/gelation

Gel point detection-Approach 2

2. G' , G'' versus time behavior



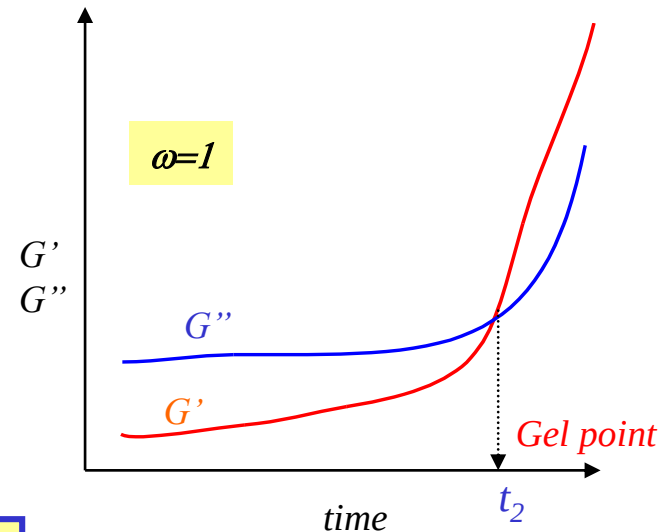
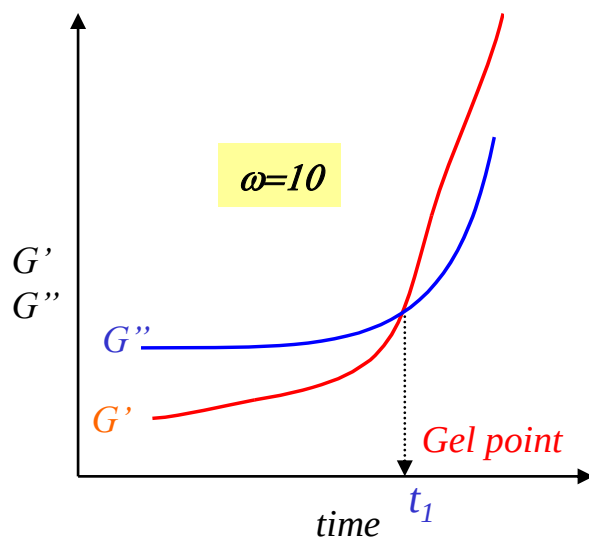
Intersection where $G'=G'' \Rightarrow$ Gel point

Problem: Do we get same point for different frequencies?

Gel point detection – Approach 2, cont'd

2. G' , G'' versus time behavior

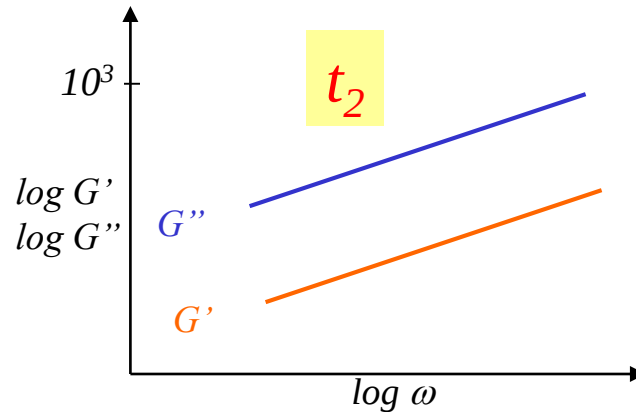
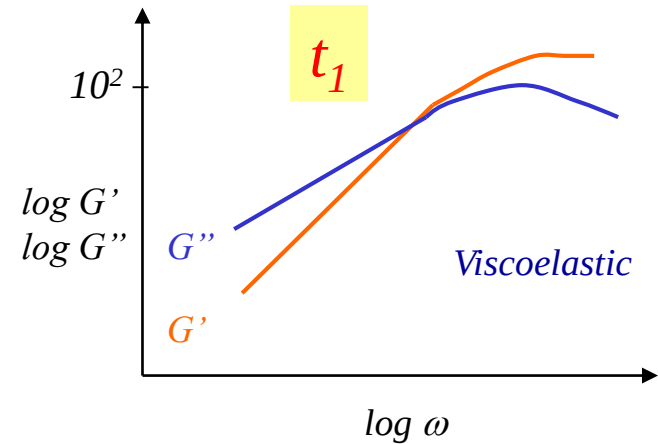
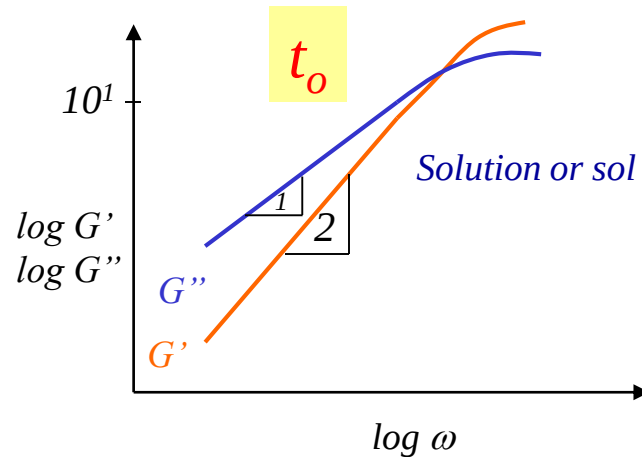
- The technique is ok if gel point is frequency independent. But often this may not be the case.



$$t_1 \neq t_2$$

Gel point detection – Approach 3

3. G' , G'' versus frequency at different time intervals



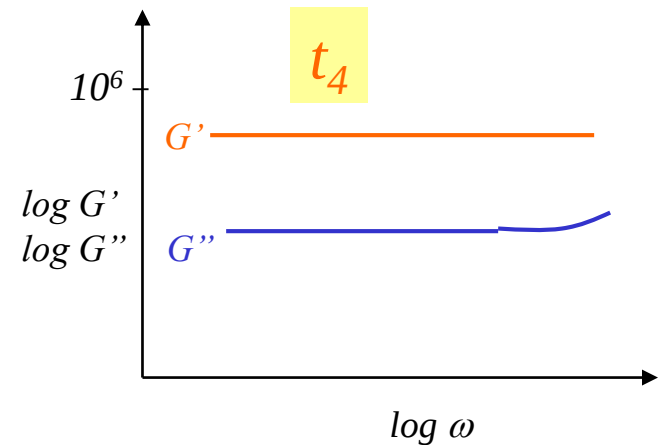
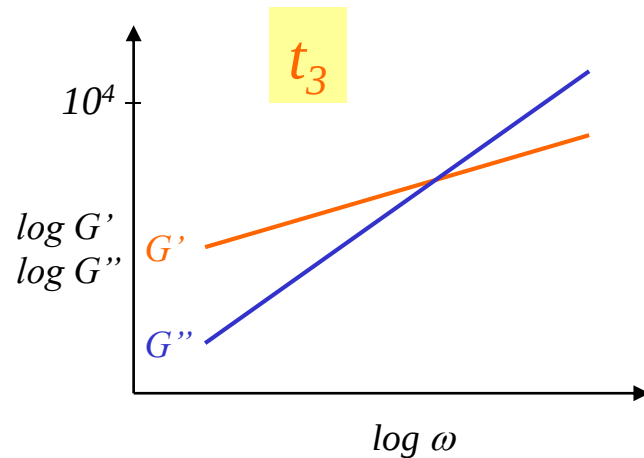
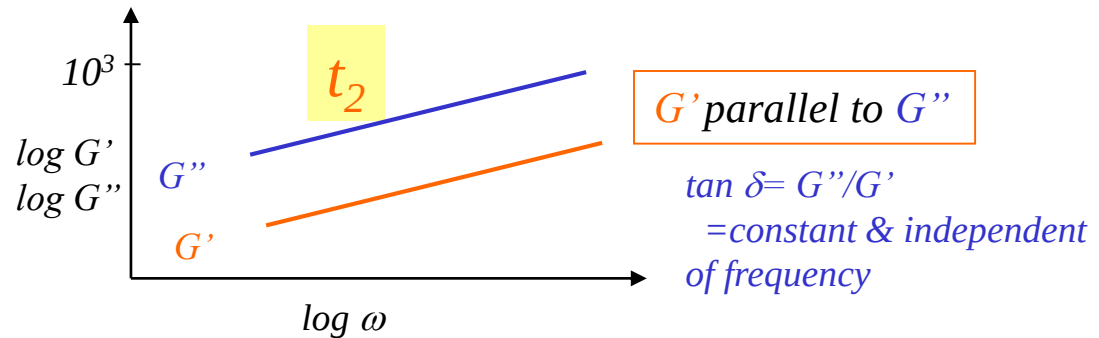
G' parallel to G''

Gel Point

Winter-Chambon
criteria

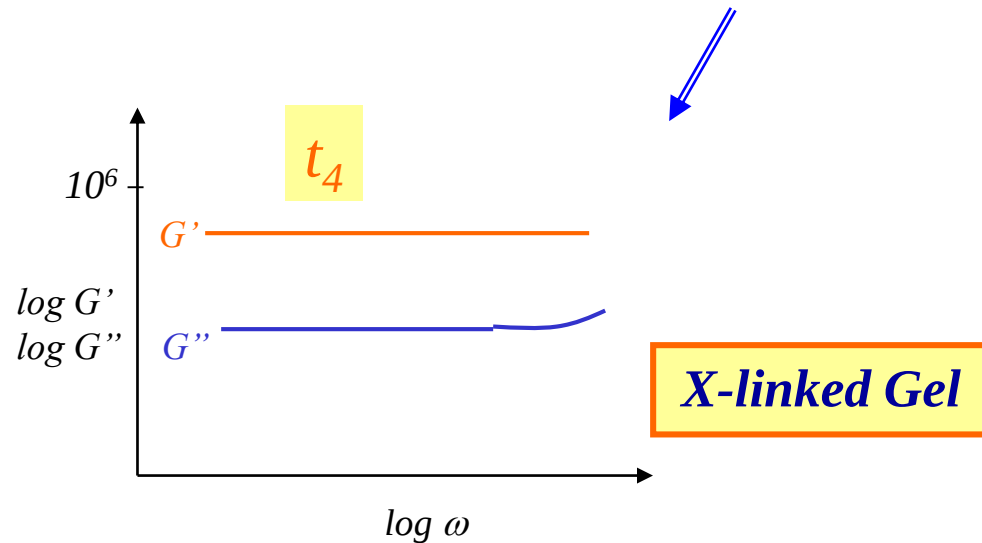
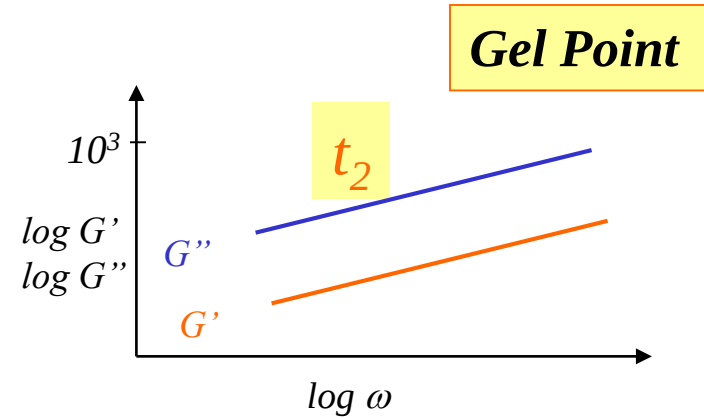
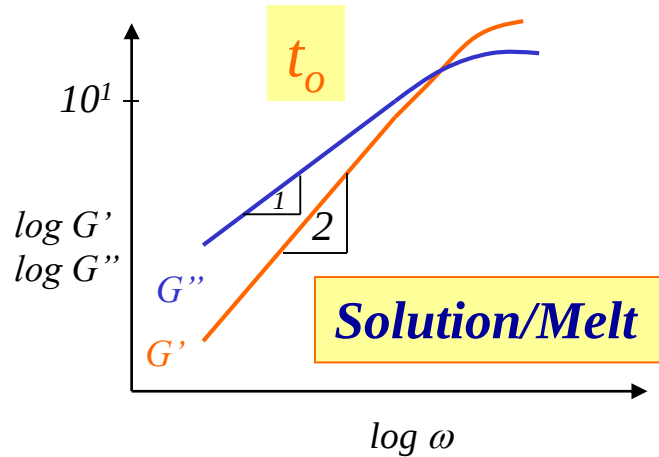
Gelation & Gel point detection

Gel Point



$G' \gg G''$
 G', G'' frequency independent
 = **Fully x-linked gel**

Recap- Gelation & Gel point detection

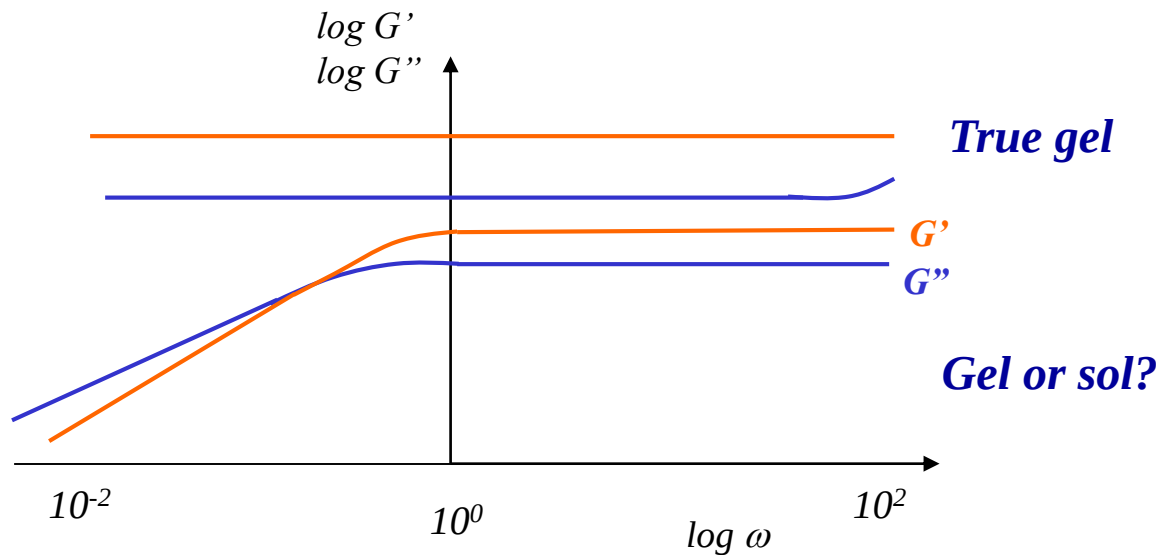


At the Gel Point

- *Molecular weight M_w goes to infinite*
- *System loses solubility*
- *Zero shear viscosity goes to infinite*
- *Equilibrium Modulus is zero and starts to rise to a finite number beyond the gel point*

Exercise – Is this a gel or solution?

Go to lower frequencies to insure G' and G'' are frequency independent



Back to Gel point analysis

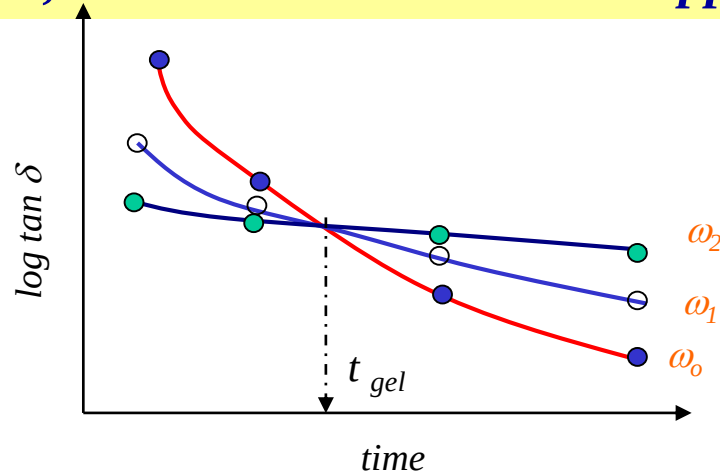
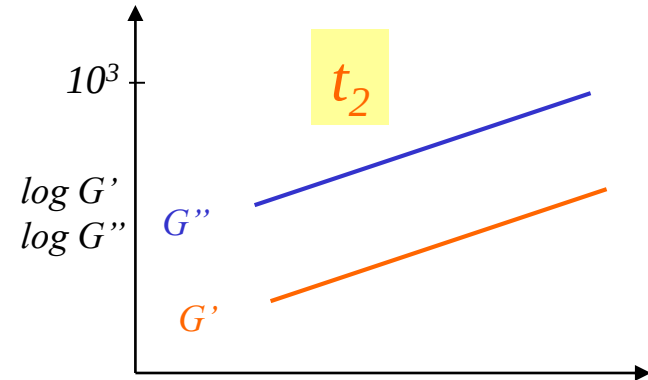
Concern:

*What if we have large enough time interval
so we miss gelation point (t_2 behavior)*

Solution:

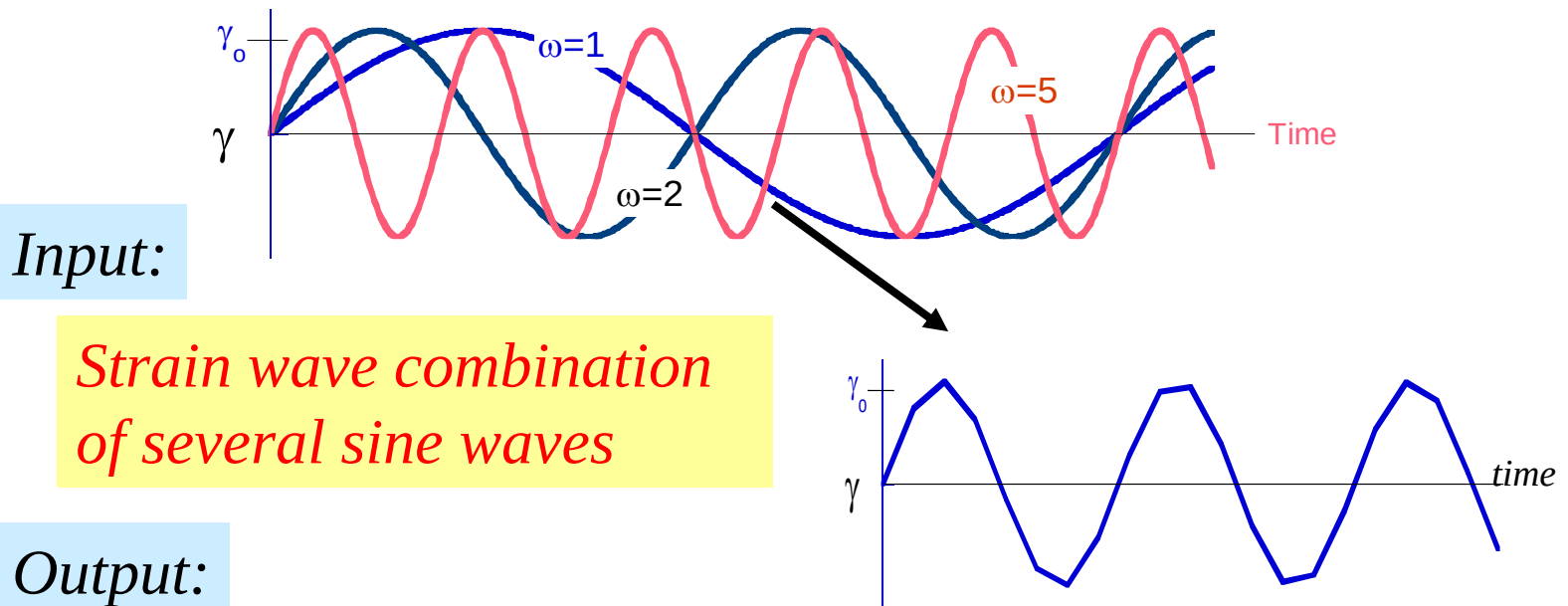
Plot $\tan \delta$ versus time

For fast reactions, in addition use Multiwave approach



Multiwave Approach

Combine several frequencies/oscillation into one.
Fourier Transform Mechanical Spectroscopy



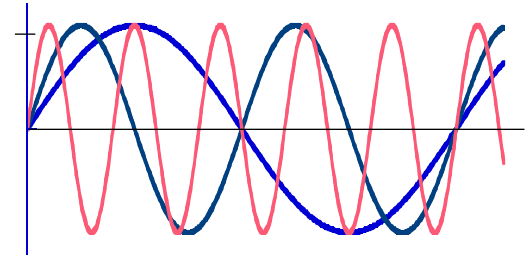
Complex torque signal $\xrightarrow{\text{Fourier transform}}$

G', G'' for several frequencies from one data point

Multiwave Approach, cont'd

Advantages:

- *Effective in fast reacting system*
- *Only one data point gives several frequencies*
 - Can get G' , G'' versus ω from 1 data point*
- *Can monitor x-linking quickly*

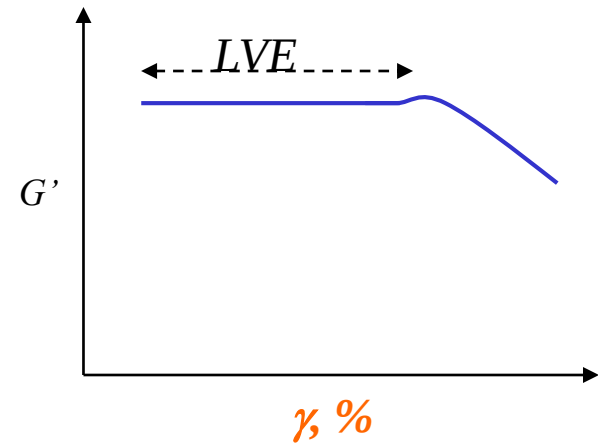


Multiwave Approach, cont'd

Pick frequency ω_1 : other frequencies have to be multiples of ω_1

Input is superposition of multiple oscillatory strains: $\gamma = \sum_{i=1}^m \gamma_i \sin \omega_i t$

Linearity condition
 $\gamma_1 + \gamma_2 + \gamma_3 + \dots + \gamma_{10} < LVE \text{ limit}$



Next Class

More on fractals – what is it?

Real data – gelation and gel point

Physical vs chemical gels

Thixotropy